



Lesson 4.4 — Graphing Rational Functions

Big idea: Sketch a rational function by finding intercepts, holes, vertical asymptotes, horizontal/slant asymptotes, and using a sign chart to handle each region in between.

Quick review

Step 1: factor numerator and denominator. Identify holes (cancellations) and VAs (remaining denominator zeros).

Step 2: find x-intercepts (numerator zeros that DON'T cancel) and the y-intercept ($f(0)$).

Step 3: end behavior — horizontal or slant asymptote (from degree comparison).

Step 4: sign chart: pick test values in each interval bounded by VAs and x-intercepts to decide whether the graph is above or below the x-axis.

Step 5: connect smoothly, respecting all asymptotes and holes.

Worked example

Worked example. Sketch $f(x) = (x^2 - 4)/(x - 1)$.

Factor: $(x - 2)(x + 2)/(x - 1)$. VA: $x = 1$. Slant: divide $\rightarrow y = x + 1 + (-3)/(x - 1)$, so $y = x + 1$.

x-intercepts: 2, -2. y-intercept: $-4/(-1) = 4$. Sign chart with breaks at $x = -2, 1, 2 \rightarrow +, -, +, +$. Sketch accordingly.

Warm-ups

1. List the asymptotes of $f(x) = 1/(x - 3)$.
2. Find the x-intercept(s) of $f(x) = (x - 2)/(x + 5)$.
3. Find the y-intercept of $f(x) = (x + 1)/(x - 4)$.

Practice

1. Sketch (describe) $f(x) = (x - 2)/(x^2 - 4)$.
2. Sketch (describe) $f(x) = (x^2 - 1)/(x^2 - 4)$.
3. Sketch (describe) $f(x) = x/(x^2 + 1)$ and find any intercept(s) and asymptotes.

Challenge

1. Sketch $f(x) = (x^2 - 1)/(x - 2)$. Identify all intercepts, asymptotes, holes, and end behavior.
2. A rational function $f(x)$ has VA at $x = 0$ and $x = 4$, hole at $x = 2$, x-intercept at $x = -1$, horizontal asymptote $y = 3$. Write a possible formula.



Answer key — Lesson 4.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. VA: $x = 3$, HA: $y = 0$.
2. $x = 2$.
3. $(0 + 1)/(0 - 4) = -1/4$.

Practice

1. Factor: $(x - 2)/[(x - 2)(x + 2)] \rightarrow$ simplified $1/(x + 2)$. Hole at $x = 2$ (height $1/4$). VA at $x = -2$. HA: $y = 0$. y-intercept $f(0) = 1/2$. No x-intercepts (after cancellation, numerator is 1).
2. $(x - 1)(x + 1)/[(x - 2)(x + 2)]$. VAs at $x = \pm 2$. HA: $y = 1$ ($n = d = 2$, ratio 1). x-intercepts: $x = \pm 1$. y-intercept: $1/4$.
3. No VA (denominator never zero). HA: $y = 0$. Intercepts: $f(0) = 0$, x-intercept $x = 0$. As $x \rightarrow \pm\infty$, $f \rightarrow 0$. Has max at $x = 1$ (value $1/2$) and min at $x = -1$ (value $-1/2$).

Challenge

1. $(x - 1)(x + 1)/(x - 2)$. VA: $x = 2$. Slant: divide $\rightarrow y = x$ with remainder 3, so $y = x + 0 + 3/(x - 2) \rightarrow$ slant $y = x$. x-intercepts: ± 1 . y-intercept: $-1/(-2) = 1/2$. Sign chart breaks at $-1, 1, 2$: signs $+, -, +, +$.
2. Need 3 in HA $\rightarrow$ degrees of P and Q match, leading ratio 3. Try $f(x) = 3(x + 1)(x - 2)/[x(x - 4)(x - 2)] \dots$ wait, that gives degree 3 in num and 3 in den. Better: $f(x) = 3(x + 1)(x - 2)/[(x)(x - 4)(x - 2)]$ — keep it cleaner: $f(x) = [3(x + 1)(x - 2)] / [x(x - 4)(x - 2) \cdot (x - 2)/(x - 2)]$ is messy. Cleanest: $f(x) = 3(x + 1)(x - 2)/[x(x - 4)(x - 2)]$; cancel $(x - 2)$: hole at 2 ✓. After cancel: $3(x + 1)/[x(x - 4)]$. That has degrees 1/2 and HA $y = 0$, NOT 3. So we need top and bottom degrees to match BEFORE cancellation. Use $f(x) = 3(x + 1)^2(x - 2) / [x(x - 4)(x - 2)]$. Cancel $(x - 2)$: $3(x + 1)^2/[x(x - 4)]$, degrees 2/2 with leading 3 $\rightarrow$ HA $y = 3$ ✓. VAs at 0, 4 ✓. Hole at 2 ✓. x-intercept at -1 ✓.